Inflation, Output and the Phillips Curve

The Early Empirical Consensus (1950s–1960s)

- A.W. Phillips (1958) documented a stable, negative empirical relationship between nominal wage growth $(g_{w,t})$ and unemployment (u_t).

$$g_{w,t} = \frac{W_t - W_{t-1}}{W_{t-1}} = f(u_t) \quad \text{where} \quad f'(u_t) < 0$$

- Samuelson and Solow (1960) modified this to link price inflation (π_t) and unemployment. To derive this, assume firms set prices (P_t) with a constant markup (μ) over unit labor costs, where A_t is labor productivity and W_t is the nominal wage:

$$P_t = (1 + \mu)\frac{W_t}{A_t}$$

  Taking natural logarithms and differentiating with respect to time (to approximate percentage changes), let a be the constant growth rate of labor productivity:

$$\pi_t = g_{w,t} - a$$

  Substituting the wage function $f(u_t)$ into the price equation yields the original Keynesian Phillips Curve:

$$\pi_t = f(u_t) - a$$

Note: Policymakers believed they could treat this equation as a menu of options, permanently choosing a lower unemployment rate at the cost of a permanently higher, but stable, inflation.

The Accelerationist Challenge and the Natural Rate (Late 1960s)

- M. Friedman and E. Phelps challenged the stable tradeoff. They argued that workers supply labor based on the *real wage*, not the nominal wage. Thus, wage negotiations will incorporate expected inflation (π_t^e). The wage adjustment equation is modified to include expectations:

$$g_{w,t} = f(u_t) + \pi_t^e$$

  Assuming a linear specification for the function $f(u_t) = -\beta(u_t - u^*)$, where u^* represents the Natural Rate of Unemployment (determined purely by real structural/frictional factors in the economy).

$$g_{w,t} = -\beta(u_t - u^*) + \pi_t^e$$

  Substituting this back into the price markup equation and assuming productivity growth is absorbed into the definition of u^*, we arrive at the foundational augmented curve.

$$\pi_t = \pi_t^e - \beta(u_t - u^*) + v_t$$

Adaptive Expectations and the NAIRU

- To make the augmented curve operational, Phelps modeled expected inflation using an adaptive expectations hypothesis. Agents update their expectations based on past forecast errors.

- The adaptive expectations rule is defined mathematically as:

$$\pi_t^e - \pi_{t-1}^e = \theta(\pi_{t-1} - \pi_{t-1}^e) \quad \text{where} \quad 0 < \theta \leq 1$$

- Rearranging this to solve for expected inflation gives a weighted average of past inflation and past expectations:

$$\pi_t^e = \theta\pi_{t-1} + (1-\theta)\pi_{t-1}^e$$

- Deriving the Accelerationist Model:
  - Assume $\theta = 1$, meaning agents fully and immediately adjust their expectations to last period's actual inflation rate. The expectations equation simplifies to:

$$\pi_t^e = \pi_{t-1}$$

  - Substitute this assumption into the Expectations-Augmented Phillips Curve (assuming $v_t = 0$ for simplicity):

$$\pi_t = \pi_{t-1} - \beta(u_t - u^*)$$

  - Subtract π_{t-1} from both sides to express the relationship in terms of the *change* in inflation:

$$\pi_t - \pi_{t-1} = -\beta(u_t - u^*)$$

Note: In this formulation, the "natural rate" u^* becomes u^n, the NAIRU.

$$\Delta\pi_t = -\beta(u_t - u^n)$$

If unemployment falls below the NAIRU ($u_t < u^n$), inflation strictly accelerate ($\Delta\pi_t > 0$). To permanently maintain unemployment below u^n, a central bank would have to generate infinite, ever-accelerating inflation, formally proving the long-run Phillips curve is vertical.

Rational Expectations and the Lucas Critique (1970s)

- R. Lucas (1976) and T. Sargent pointed out that adaptive expectations are systematically flawed. If inflation is constantly accelerating, an adaptive rule ($\pi^e = \pi_{t-1}$) implies agents will continually and predictably under-predict inflation, which is irrational.
- Agents form expectations using all available information (Ω_{t-1}), including the central bank's policy rules and the true structural model of the economy:

$$\pi_t^e = E_{t-1}[\pi_t|\Omega_{t-1}]$$

- Mathematical Implication:
  - Under REH, the forecast error ($\pi_t - \pi_t^e$) cannot contain any predictable pattern. Actual inflation only deviates from expected inflation by a purely random, white-noise shock ε_t:

$$\pi_t - E_{t-1}[\pi_t] = \epsilon_t$$

  - Substituting this directly into the Expectations-Augmented Phillips curve:

$$\epsilon_t = -\beta(u_t - u^*)$$

  - Solving for unemployment (u_t):

$$u_t = u^* - \frac{1}{\beta}\epsilon_t$$

Note: Under rational expectations, systematic (predictable) monetary policy is completely ineffective at manipulating output or employment. Because agents perfectly anticipate

systematic policy, π_t^e adjusts instantly. Unemployment only deviates from the natural rate due to completely random, unanticipated shocks (ϵ_t).

Disinflation and Neo-Keynesian Rigidities (1980s Onward)

- The Rational Expectations framework implied that if a central bank announces a perfectly credible plan to lower inflation, $E_{t-1}[\pi_t]$ will drop immediately, allowing for a costless disinflation (no rise in unemployment).
- However, when Fed Chairman Paul Volcker instituted tight money in 1979 to break stagflation, unemployment did rise significantly.
- While the "sacrifice ratio" (the loss in output/employment per 1% drop in inflation) was lower than adaptive models predicted—proving credibility mattered—it was not zero.
- This empirical reality gave rise to New Keynesian Macroeconomics, which accepts rational expectations but introduces micro-founded nominal frictions (like Calvo staggered pricing or menu costs) to mathematically explain why prices and expectations do not adjust perfectly and instantaneously.